Analysis

PENG

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Contents

1 Introduction **1**

2 Natural Number **1**
2.1 Addition 2
2.2 Multiplication 3

3 Set Theory **3**
3.1 Cartesian Product 4

4 Function **4**
4.1 Images 5

5 Cardinality of Sets **5**

6 Integers **6**

7 Rationals **6**

8 Real Numbers **6**

9 Limits of Sequences **6**

10 Power Series **6**

11 Infinite Sets **6**

12 Continuous Functions **6**

13 Differentiation of Functions **6**

14 Riemann Integral **6**

A Tools **6**

1 Introduction

Definition 1.1 (analysis)**.** *Analysis of real numbers, sequences and series of real numbers, real-valued functions, and geometry [1, 2].*

2 Natural Number

Definition 2.1 (natural number)**.** *A natural number is any element of the set*

$$\mathbb{N} := 0, 1, 2, 3, \ldots \tag{2.1}$$

Axiom 2.1 (0)**.** *0 is a natural number.*

Axiom 2.2 (successor). *If n is a natural number, then $n{+}{+}$ is also a natural number.*

Definition 2.2 (notation). $1 := 0{+}{+}$, $2 := 1{+}{+}$, $3 := 2{+}{+}$, ...

Axiom 2.3 ($n{+}{+} \neq 0$). *0 is not the successor of any natural number n.*

Axiom 2.4 ($n{+}{+} \neq m{+}{+}$). *Different natural numbers must have different successors. If $n \neq m$, then $n{+}{+} \neq m{+}{+}$. Equivalently, if $n{+}{+} = m{+}{+}$, then $n = m$.*

Axiom 2.5 (principle of mathematical induction). *Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{+}{+})$ is also true. Then $P(n)$ is true for every natural number n.*

Lemma 2.1 (recursive definition of sequence). *$a_0 = c$ and $a_{n{+}{+}=f_n(a_n)}$ for each natural number n.*

2.1 Addition

Definition 2.3 (addition). *Define $0 + m := m$. Suppose inductively that we have defined $n + m$. Then $(n{+}{+}) + m := (n + m){+}{+}$.*

$a_n = n + m$ and $f_n(a_n) = a_n{+}{+}$

Lemma 2.2. $n + 0 = n$

Lemma 2.3. $n + (m{+}{+}) = (n + m){+}{+}$

Proposition 2.1 (addition is commutative).

$$n + m = m + n \tag{2.2}$$

Proposition 2.2 (addition is associative).

$$(a + b) + c = a + (b + c) \tag{2.3}$$

Proposition 2.3 (cancellation law). *If $a + b = a + c$, then $b = c$.*

Definition 2.4. *A natural number n is positive if and only if $n \neq 0$.*

Lemma 2.4. *If a is a positive natural number and b is a natural number, then $a + b$ is positive.*

Lemma 2.5. *If $a + b = 0$, then $a = 0$ and $b = 0$.*

Lemma 2.6. *If a is a positive number, then there exists exactly one natural number b such that $b{+}{+} = a$.*

Definition 2.5 (order).

- *$n \geq m$ or $m \leq n$ if and only if $n = m + a$ for some natural number a*
- *$n > m$ or $m < n$ if and only if $n \geq m$ and $n \neq m$*

Proposition 2.4 (properties of order).

- *reflexive: $a \geq a$*
- *transitive: If $a \geq b$ and $b \geq c$, then $a \geq c$*
- *antisymmetric: If $a \geq b$ and $b \geq a$, then $a = b$*
- *addition preserve order: $a \geq b$ if and only if $a + c \geq b + c$*
- *$a < b$ if and only if $a{+}{+} \leq b$*
- *$a < b$ if and only if $b = a + d$ for some positive number d*

Proposition 2.5 (trichotomy of order). *Exactly one of following statement is true: $a < b$, $a = b$, or $a > b$.*

Proposition 2.6 (strong principle of induction). *Let m_0 be a natural number. Suppose that for each $m \geq m_0$, if $P(m')$ is true for all $m_0 \leq m' < m$, then $P(m)$ is also true. Then $P(m)$ is true for all $m \geq m_0$.*

Example 2.1 (Fibonacci sequence).

$$F_0 = 0 \tag{2.4}$$

$$F_1 = 1 \tag{2.5}$$

$$F_n = F_{n-1} + F_{n-2} \tag{2.6}$$

2.2 Multiplication

Definition 2.6 (multiplication). *Define* $0 \times m := 0$. *Suppose inductively that we have defined* $n \times m$. *Then* $(n++) \times m := (n \times m) + m$.

Lemma 2.7. $n \times 0 = 0 \times n$

Lemma 2.8. $n \times (m++) = (n \times m) + n$

Proposition 2.7 (multiplication is commutative).

$$n \times m = m \times n \tag{2.7}$$

Lemma 2.9 (zero divisor). $n \times m = 0$ *if and only if at least one of* n *and* m *is equal to 0. If* n *and* m *are both positive, then* nm *is also positive.*

Proposition 2.8 (distributive law).

$$a(b+c) = (b+c)a = bc + ca \tag{2.8}$$

Proposition 2.9 (multiplication is associative).

$$(ab)c = a(bc) \tag{2.9}$$

Lemma 2.10 (multiplication preserve order). *If* $a < b$ *and* c *is positive, then* $ac < nc$.

Proposition 2.10 (cancellation law). *If* $ac = bc$ *and* c *is positive, then* $a = b$.

Proposition 2.11 (Euclid's division lemma). *Let* n *be a natural number and* q *be a positive natural number. Then there exist natural numbers* m *and* r, *such that* $0 \leq r < q$ *and* $n = mq + r$.

Proof. $0 = 0 \times q + 0$ and $0 \leq r < q$. Suppose that $n = mq + r$ and $0 \leq r < q$. Then $n++ = mq + (r++)$ and $0 \leq r++ \leq q$. If $r++ = q$, $n++ = mq + q = (m+1)q + 0$. If $r++ < q$, $n++ = mq + r++$. □

Definition 2.7 (Exponentiation). *Define* $m^0 := 1$, *in particular* $0^0 := 1$. *Suppose recursively that* m^n *has been defined for some natural number* n, *then define* $m^{n++} := m^n \times m$.

3 Set Theory

Definition 3.1 (set). *Define a set* A *to be any unordered collection of objects. If* x *is an element of* A, *then* $x \in A$, *otherwise* $x \notin A$.

Axiom 3.1 (sets are objects). *If* A *is a set, then* A *is also an object.*

Axiom 3.2 (equality of sets). $A = B$ *if and only if every element of* A *is an element of* B *and vice versa.*

Axiom 3.3 (empty set). *There exists an empty set* $\emptyset$ *which contains no elements.*

Lemma 3.1 (single choice). *If* A *is a non-empty set, then there exists an object* x *such that* $x \in A$.

Axiom 3.4 (singleton and pair sets). *If* a *is an object, then there exists a singleton set* $\{a\}$ *whose only element is* a. *If* a *and* b *are objects, then there exists a pair set* $\{a, b\}$ *whose only elements are* a *and* b.

Axiom 3.5 (pairwise union). *If* A *and* B *are two sets, then there exists a union of* A *and* B *set* $A \cup B$, *which consists of all the elements which belong to* A *or* B *or both.*

$$x \in A \cup B \iff x \in A \text{ or } x \in B \tag{3.1}$$

Proposition 3.1 (properties of union).

- $\{a, b\} = \{a\} \cup \{b\}$
- *commutative:* $A \cup B = B \cup A$
- *associative:* $(A \cup B) \cup C = A \cup (B \cup C)$

Definition 3.2 (subsets). *A is a subset of B, denoted $A \subseteq B$, if and only if every element of A is also an element of B.*

A is a proper subset of B, denoted $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Axiom 3.6 (axiom of specification). *Let A be a set and $P(x)$ be a property pertaining to x for each $x \in A$. Then there exists a set $\{x \in A : P(x) \text{ is true}\}$ whose elements are precisely the elements x in A for which $P(x)$ is true.*

Definition 3.3 (intersections). *The intersection of two sets is defined to be the set*

$$A \cap B := \{x \in A : x \in B\} \tag{3.2}$$

Definition 3.4 (difference sets). *Define $A - B$ or $A \backslash B$ to be set A with elements of B removed*

$$A \backslash B := \{x \in A : x \notin B\} \tag{3.3}$$

Axiom 3.7 (replacement). *If $P(x, y)$ is a property pertaining to x and y, such that for each $x \in A$ there is at most one y for which $P(x, y)$ is true. Then there exists a set $\{y : P(x, y) \text{ is true for some } x \in A\}$.*

Axiom 3.8 (infinity). *There exists a set $\mathbb{N}$, whose elements are called natural numbers, as well as an object 0 in $\mathbb{N}$, and an object $n{+}{+}$ assigned to every $n \in \mathbb{N}$, such that the Peano axioms hold.*

3.1 Cartesian Product

Definition 3.5 (ordered pair). *If x and y are any objects, define the ordered pair (x, y) to be a new object. Two ordered pairs (x, y) and (x', y') are equal if and only if both their components match*

$$(x, y) = (x', y') \iff x = x' \text{ and } y = y'. \tag{3.4}$$

Definition 3.6 (Cartesian product). *If X and Y are sets, then define Cartesian product $X \times Y$ to be the collection of ordered pair*

$$X \times Y = \{(x, y) : x \in X, y \in Y\} \tag{3.5}$$

Definition 3.7 (ordered sequence of n elements). *If $(x_1, \ldots, x_n)$ is an ordered n-tuple, define Cartesian product $X_1 \times \cdots \times X_n$ by*

$$\prod_{1 \leq i \leq n} X_i := \{(x_i)_{1 \leq i \leq n} : x_i \in X_i \text{ for all } 1 \leq i \leq n\} \tag{3.6}$$

Lemma 3.2 (finite choice). *Let X_i be a non-empty set, then there exists an n-tuple $(x_i)_{1 \leq i \leq n}$ such that $x_i \in X_i$ for all $1 \leq i \leq n$.*

4 Function

Definition 4.1 (function). *Let X, Y be sets, and let $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$, such that for every $x \in X$, there is exactly only one $y \in Y$ for which $P(x, y)$ is true. Then the function $f : X \to Y$ is defined by P on the domain X and codomain (range) to be the unique object $f(x) \in Y$ for which $P(x, f(x))$ is true.*

Definition 4.2 (equality of functions). *Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if their domains and codomains agree, and $f(x) = g(x)$ for all $x \in X$.*

Definition 4.3 (composition). *Let $f : X \to Y$ and $g : Y \to Z$ be two functions, such that the codomain of f is the same set as the domain of g. Then define composition $g \circ f : X \to Z$ to the function*

$$(g \circ f)(x) := g(f(x)) \tag{4.1}$$

Definition 4.4 (injective). *A function is one-to-one if every element in X maps to different element in Y*

$$x \neq x' \implies f(x) \neq f(x') \tag{4.2}$$

Equivalently,

$$f(x) = f(x') \implies x = x' \tag{4.3}$$

Definition 4.5 (surjective). *A function is onto if every element in Y comes from applying f to some element in X.*

$$\text{For every } y \in Y, \text{ there exists } x \in X \text{ such that } f(x) = y. \tag{4.4}$$

Definition 4.6 (bijective). *A function is bijective or invertible if it is both one-to-one and onto.*

4.1 Images

Definition 4.7 (images of sets)**.** *If $f : X \to Y$ is a function from X to Y, and S is a subset of X, define $f(S)$ to be the set*

$$f(S) := \{f(x) : x \in S\} \tag{4.5}$$

Definition 4.8 (inverse images)**.** *If U is a subset of Y, define the set $f^{-1}(U)$ to be the set*

$$f^{-1}(U) := \{x \in X : f(x) \in U\} \tag{4.6}$$

Axiom 4.1 (power set axiom)**.** *Let X and Y be sets. Then there exists a set Y^X which consists of all the functions from X to Y*

$$f \in Y^X \iff f \text{ is a function with domain } X \text{ and codomain } Y. \tag{4.7}$$

Axiom 4.2 (union)**.** *Let A be a set whose elements are sets. Then there exists a set $\bigcup A$*

$$x \in \bigcup A \iff x \in S \text{ for some } S \in A. \tag{4.8}$$

5 Cardinality of Sets

Definition 5.1 (ordinal and cardinal)**.** *The cardinals are One, Two, ..., and are used to count how many things there are in a set.*

THe ordinals are First, Second, ..., and are used to order a sequence of objects.

Natural numbers can be used to count the cardinality of sets, as long as the set is finite.

Definition 5.2 (equal cardinality)**.** *X and Y have equal cardinality if and only if there exists a bijection $f : X \to Y$ from X to Y.*

Proposition 5.1 (properties of equal cardinality)**.**

- *X has equal cardinality with X.*
- *If X has equal cardinality with Y, then Y has equal cardinality with X.*
- *If X has equal cardinality with Y and Y has equal cardinality with Z, then X has equal cardinality with Z.*

Definition 5.3 (cardinality n)**.** *A set X has cardinality n or n elements if and only if it has equality cardinality with $\{i \in \mathbb{N} : 1 \leq i \leq n\}$.*

Proposition 5.2 (uniqueness of cardinality)**.**

Definition 5.4 (finite sets)**.**

Theorem 5.1 ($\mathbb{N}$ is infinite)**.** *The set of natural number $\mathbb{N}$ is an infinite set.*

Proof. □

Proposition 5.3 (cardinal arithmetic)**.**

6 Integers

7 Rationals

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A Tools

Definition A.1 (implication)**.** *"If P, then Q" = "If not Q, then not P".*

Lemma A.1 (de mongan laws)**.**

- *not "A and B" = "not A or not B"*
- *not "A or B" = "not A and not B"*

Axiom A.1 (subsititution axiom)**.** *Given any two objects x and y, if $x = y$, then $f(x) = f(y)$ for all functions or operations. For any property $P(x)$ pertaining to x, if $x = y$, then $P(x)$ and $P(y)$ are equivalent statements.*

Definition A.2 (conjunction and disjunction)**.**

- *The statement "X and Y" is true if X and Y are both true.*
- *The statement "X or Y" is true if either X or Y is true, or both.*

References

[1] Terence Tao. *Analysis I.* Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.

[2] Terence Tao. *Analysis II.* Vol. 2. Springer, 2022. ISBN: 978-981-19-7284-3.